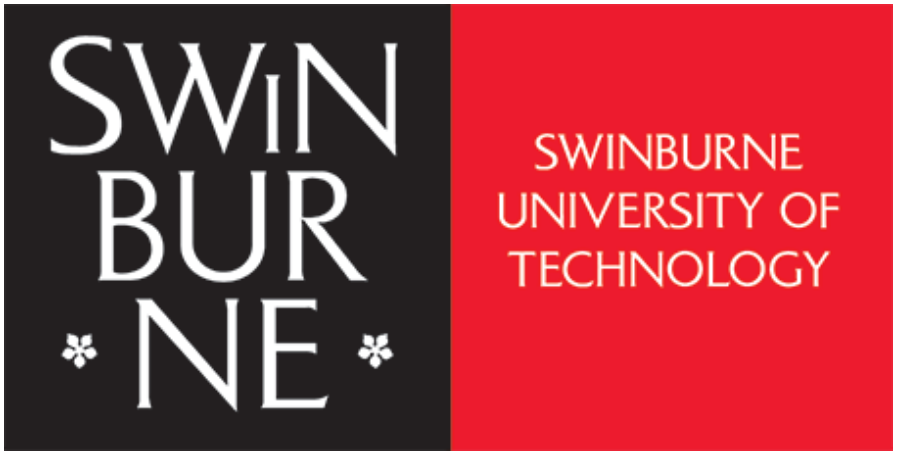


2025-COS30049-Computing Technology Innovation Project

Workshop Guide

Workshop 04

Machine Learning Project Lifecycle Introduction & Data Processing via Python

Objective:

By the end of this workshop, participants will be able to set up their Python environment in Visual Studio Code and apply core data preprocessing techniques. The focus is on practical skills for preparing security-related datasets, including basic cleaning, handling missing values, and extracting useful features for later machine learning tasks.

Learning Outcomes:

1. **Setting Up Python Environment:** Successfully configure and run Python with Jupyter notebooks in Visual Studio Code.

2. **Handling Missing Data:** Learn practical techniques to detect and manage missing values in datasets.

3. **Data Cleaning & Feature Preparation:** Gain experience in cleaning raw data and extracting useful features from security logs for later ML tasks.

Hands-on Practice:

Participants will complete step-by-step exercises using synthetic security logs to practice preprocessing, including cleaning, handling missing data, and preparing features. This hands-on experience will build confidence in managing real-world cybersecurity datasets.

Workshop Structure:

1. Configuration and Setup (10 mins)

Please refer to another file from Canvas under this week “Comprehensive Guide_ Installing Python, Setting Up VS Code, and Configuring the System Environment Path.pdf”.

2. Introduction to pip (15 mins)

pip, a powerful package management tool for Python. pip allows you to easily install, update, and manage Python libraries and dependencies, which are essential for data processing and machine learning projects.

I. **Installing pip (you should already be installed after section 1):**
   - Instructions on how to check if pip is installed:
     pip –version

II. **Basic pip Commands:**
   - Installing a package:
     pip install package_name
   - Upgrading a package:
     pip install --upgrade package_name
   - Listing installed packages:
     pip list
   - Uninstalling a package:
     pip uninstall package_name

3. Swap to the .ipynb file to continue the workshop (80 mins)

   Go to the Canvas, Modules → Week4 → download the .ipynb file to continue the workshop.

4. Reflection and Closing (5 mins):

   Allow students to share what they learned, found interesting, or had difficulty understanding.